

Group Consensus and Outputs so far:

1)

Stochastic Differential Equations

$$dC_i = (\alpha_i C_i + \sum_{j \in B(i)} \frac{1}{|B(i)|} \alpha_j C_j - \beta_i C_i (P_i(t))^{-d}) dt + C_i \sigma_i dW_i$$

$$E_i = \lambda_i C_i$$

$$dE_i = (\alpha_i E_i + \sum_{j \in B(i)} \frac{1}{|B(i)|} \alpha_j E_j - \beta_i E_i (P_i(t))^{-d}) dt + E_i \sigma_i dW_i$$

($E_i = E_i(t)$ short hand notation)

2)

Prophet and SARIMA Models

3)

2021-2025

4)

Sexual Offenses, Antisocial Behavior

5)

$d = 0.3$ (maybe we find more specific literature that is about Sexual Offenses, Antisocial Behavior)

6)

Coding of the SDEs

7)

Visualizations of the SDEs

8)

Finding $\alpha_i, \alpha_j, \beta_i > 0$ via Non-Negative-Coefficients-Constrained-Least-Squares to solve Simultaneous Equations (assuming that $\sigma_i = 0$ while calculating $\alpha_i, \alpha_j, \beta_i > 0$) that satisfy known data points for $E_i(m)$ and then optimizing for σ_i such that we are closest as we can be to the known data points known data points for $E_i(m)$.

9) Let me know if i am missing anything or if you have new ideas

Discussion:

I would say lets move on with this dE_i and not work on C_i because we do not have much data on it and we do not have a strong reason to say $E_i = \lambda_i C_i$. We can simply model E_i instead of C_i and minimize E_i or extreme changes in dE_i by optimizing the data points of number of police that we used to interpolate and get $P_i(t)$. This way

redistribution or reallocation can be a policy or advice to the police or government that can easily be translated to our model and tested its effect.

Incorporating Prophet and Giving Advices: Answering the Research Question

(We need to agree on clusters of the Prophet, I think clusters and regions of the SDEs should match, otherwise how are we going to use the forecasted data in the 43-regional SDE?)

We can use Prophet to have regional forecasted data of the first half of 2026 or full 2026 (possibly 2027 either but let me know what you think) and find coefficients again accordingly. We can then simulate starting from the half 2026 or the start of 2027 which allows close future advices (giving advice for 2025 and testing it in our simulations is not something the police can use for 2026).

After observing the visualizations we can optimize for $P_i(t)$ (changing data points of number of police) such that: (in most simulations possible)

1) Optimizing distribution of police:

We could say that we are optimizing police recruitments such that we have the optimal reallocation of the number of police.

1A) minimize $\sum_{i=1}^{N_R} E_i$, or minimize E_i for some i

1B) minimize extreme changes in dE_i (maybe $\frac{dE_i}{dt}$ is a better way to see it)

1C) Setting goals like $\max\{E_i\} = M$ (maybe min also?) and no region should have more than M crimes committed monthly

1D) your ideas

2) Increasing the number of police in the regions that decrease crimes the most: In which region(s) should the government spend more money if they want to increase the budget on policing? (Spending money such that crimes are decreased the most, increasing the number of police in a region such that we decrease crimes (either whole UK or ...) the most) (It is nice if we express number of police in terms of money)

2A) minimize $\sum_{i=1}^{N_R} E_i$, or minimize E_i for some i

2B) minimize extreme changes in dE_i (maybe $\frac{dE_i}{dt}$ is a better way to see it)

2C) Setting goals like $\max\{E_t\} = M$ and no region should have more than M crimes committed monthly

3) Reducing redundant police so that the government wastes less money: We can spot the regions where decreasing the number of police makes the least effect on the number of crimes committed and (I would be surprised crimes committed in a month would decrease) minimize the number of police without increasing the crimes much so that the government spends less money. (in some regions or in whole UK) (Open for Ethical Considerations)

3A) minimize $\sum_{i=1}^{N_R} E_i$, or minimize E_i for some i

3B) minimize extreme changes in dE_i (maybe $\frac{dE_i}{dt}$ is a better way to see it)

3C) Setting goals like $\max\{E_t\} = M$ and no region should have more than M crimes committed monthly

Of course we need to show our results via simulations.

It is nice if we can find how much policing costs so that we can express this advice with money rather than the number of police.

4) Your ideas

My Ideas on Improving the Model:

New: Correlations between regions:

$$dE_i = (\alpha_i E_i + \sum_{j \in B(i)} p(i, j) \alpha_j E_j - \beta_i E_i (P_i(t))^{-d}) dt + E_i \sigma_i dW_i$$

$p(i, j)$: Correlation coefficient between i -th and j -th regions

Or even: (depending on correlations we find):

$$dE_i = (\alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} p(i, j) \alpha_j E_j - \beta_i E_i (P_i(t))^{-d}) dt + E_i \sigma_i dW_i$$

New: Capturing seasonality trends:

Assuming $t = 1$ to $t = 2$ is one monthly change:

We have peaks in summer for both Antisocial behavior and Sexual Offenses.

Monthly change could be $t = 0$ to $t = \frac{\pi}{12}$ and then $t = \frac{\pi}{2}k$, k is integer, could correspond to peaks in June.

$$dE_i = (\alpha_i E_i (1 + \gamma_i \sin(t)) + \sum_{j \in B(i)} p(i, j) \alpha_j E_j - \beta_i E_i (P_i(t))^{-d}) dt + E_i \sigma_i dW_i$$

$\gamma_i = \text{Average}[E_i(m_{a+5}) - E_i(m_a)]$? (you get what I mean)

Our $E_i(m_a)$ already counts both types of crimes. Maybe it should be a yearly function?

Reasoning: (from Sven's SSA3)

